# Structured notes: Module II: free electron theory and band theory

Puri and Babbar, *Solid State Physics and Electronics*, chapters 5 and 6 (printed pages 151–198). Text extracted by Codex from the page images and checked against them. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Problems are not included.

## Section 1: Drude–Lorentz classical theory (Puri and Babbar 5.1)

### 5.1 Drude–Lorentz classical free electron theory (structured notes)

*Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 151–152. Figure 5.1 is redrawn.*

#### Before you start

- Current density $J = nev_d$; Ohm's law (Plus Two recall below).
- Maxwell–Boltzmann statistics as a classical distribution.

> **PLUS TWO RECALL: CURRENT AND OHM'S LAW**
>
> **Recall.** $I = neAv_d$ and $V = IR$.
>
> **Re-derive.**
>
> 1. $J = I/A = nev_d$.
> 2. With $E = V/\ell$ and drift proportional to $E$, $J = \sigma E$.
>
> **Saved.** $J = \sigma E$.

#### 5.1.1 Assumptions

Metal = positive ion cores + valence electrons. The ion-core potential is constant; electron–electron repulsion is neglected; the electrons form a free electron gas with the properties of a perfect gas. The gas is negatively charged and dense, and the electrons obey the Pauli principle. The potential energy inside the metal is zero, so the total energy is kinetic. A barrier $V_0$ at the surface confines the electrons (Fig. 5.1).

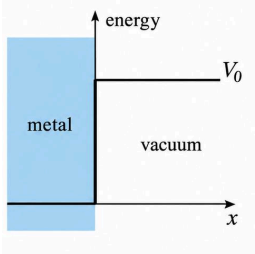



*Fig. 5.1 (book): metallic surface bounded by the potential barrier $V_0$, the difference in potential energy of an electron inside the metal and just outside it.*

#### 5.1.2 Statistics

Drude's perfect-gas picture and Lorentz's Maxwell–Boltzmann assumption (1909) define the Drude–Lorentz theory; the classical statistics give the name.

#### 5.1.3 Successes

1. **Ohm's law.** Random motion gives no current; a field gives drift, limited by collisions with the ions. The steady current is proportional to $V$ at constant $T$.
2. **High conductivity** from free motion, and the Wiedemann–Franz ratio $\sigma/\kappa$ is the same for all metals at a given $T$ (realised in practice).
3. **Lustre and opacity** from absorption and re-emission by free electrons.

#### 5.1.4 Failures

1. Resistivity predicted to vary as $T^{1/2}$; observed linear in $T$.
2. Heat capacity and paramagnetic susceptibility of conduction electrons too large (classical statistics give every electron energy).

3. Long mean free paths at low $T$ are not explained.

Fermi–Dirac statistics (Section 5.2) explain items 2 and 3 for the stated reason: only a fraction of the electrons can be excited.

**RESULT**

The classical theory gives Ohm's law and the Wiedemann–Franz ratio, and fails on heat capacity, susceptibility and the $T$-dependence of resistivity.

### Summary

- Potential box of depth $V_0$; Maxwell–Boltzmann statistics; successes and failures as above.

## Section 2: Sommerfeld theory, one-dimensional box (5.2, 5.2.1)

## 5.2 Sommerfeld's quantum theory: the one-dimensional free electron gas (structured notes)

*Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 153–159. Equations 5.1 to 5.18 are cited by their book numbers. Figures are redrawn.*

### Before you start

- Standing waves between fixed ends (box below).
- Pauli exclusion principle; two spin states per spatial orbital.

**PLUS TWO RECALL: STANDING WAVES BETWEEN FIXED ENDS**

**Recall.** Standing waves on a string of length $L$ with fixed ends have $\lambda_n = 2L/n$.

**Re-derive.**

1. Displacement $\propto \sin(n\pi x/L)$ satisfies the end conditions.
2. $k = n\pi/L$; with $p = \hbar k$ and $E = p^2/2m$, $E_n = \hbar^2 n^2 \pi^2 / 2mL^2$.

**Saved.** Eq. (5.10).

### 5.2.1 The box and its wave functions

$$V(x) = 0 \text{ for } 0 < x < L$$
$$V(x) = \infty \text{ for } x \le 0 \text{ and } x \ge L \tag{5.1}$$

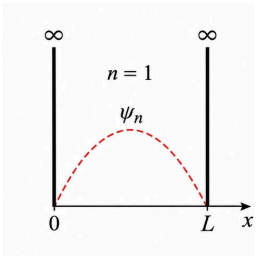



*Fig. 5.2 (book): one-dimensional potential box bounded by infinite potential energy barriers.*

Inside the box, (5.2) reduces to (5.3), with general solution (5.4) and $k$ from (5.5). The conditions (5.6) follow from the infinite walls: $\psi_n(0) = \psi_n(L) = 0$. They give $B = 0$ (5.7) and $k = n\pi/L$ (5.8), so $\psi_n = A \sin(n\pi x/L)$ (5.9) and

$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2 = \frac{n^2 h^2}{8mL^2} \tag{5.10}$$

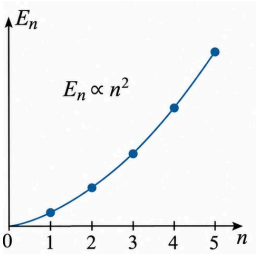



*Fig. 5.3 (book): $E_n$ versus n for a one-dimensional crystal.*

The quantum number $n$ takes integer values. The spacing shrinks as $L$ grows (Fig. 5.3).

Normalisation: $A^2 \int_0^L \sin^2(n\pi x/L)\,dx = 1$ gives $A = \sqrt{2/L}$ and

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right) \tag{5.11}$$

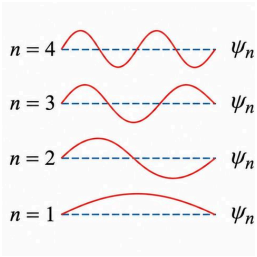



*Fig. 5.4 (book): first four wave functions (solid lines) and the corresponding energy levels (broken lines).*

### 5.2.2 Fermi energy

Two electrons per level; $N$ electrons fill levels up to $n_F = N/2$ (5.12). The Fermi energy (5.13):

$$E_F = \frac{\hbar^2}{2m}\left(\frac{N\pi}{2L}\right)^2 \tag{5.13}$$

Worked example in the book: $N/L = 0.5$ Å$^{-1}$ gives $E_F = 2.4$ eV.

### 5.2.3 Total energy

$$E_0 = 2\sum_{n=1}^{N/2} E_n = 2 \cdot \frac{\hbar^2}{2m}\left(\frac{\pi}{L}\right)^2 \sum_{n=1}^{N/2} n^2 \tag{5.14}$$

With $\sum n^2 \simeq \frac{1}{3}(N/2)^3$, $E_0 = \frac{1}{3}NE_F$ (5.15). The average kinetic energy is one third of $E_F$.

### 5.2.4 Density of states

$D(E) = dn/dE$ (5.16) with the factor 2 for spin (5.17). Using $dE/dn = \hbar^2 n/4mL^2$ and eliminating $n$:

$$D(E) = \frac{4L}{h}\left(\frac{m}{2E}\right)^{1/2} \tag{5.18}$$

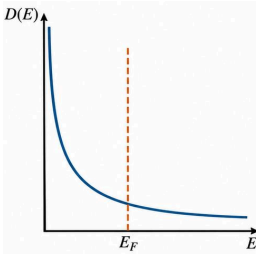



*Fig. 5.5 (book): density of states against energy for a one-dimensional metallic crystal; all states up to the Fermi level are filled at 0 K.*

> **RESULT**
>
> 1D: $E_n \propto n^2$; $E_0 = \frac{1}{3}NE_F$; $D(E) \propto E^{-1/2}$; all states below $E_F$ filled at 0 K.

## Summary

- (5.10) levels, (5.11) waves, (5.13) Fermi energy, (5.15) average energy, (5.18) density of states.

## Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}(E_n - V)\psi_n = 0 \tag{5.2}$$

$$\frac{d^2\psi_n}{dx^2} + \frac{2m}{\hbar^2}E_n\psi_n = 0 \tag{5.3}$$

$$\psi_n(x) = A\sin kx + B\cos kx \tag{5.4}$$

$$k = \sqrt{\frac{2mE_n}{\hbar^2}} \tag{5.5}$$

$$\psi_n(0) = 0 \quad \text{and} \quad \psi_n(L) = 0 \tag{5.6}$$

$$\psi_n(x) = A\sin kx \tag{5.7}$$

$$k = \frac{n\pi}{L} \tag{5.8}$$

$$\psi_n(x) = A\sin\left(\frac{n\pi}{L}\right)x \tag{5.9}$$

$$2n_F = N \tag{5.12}$$

$$E_0 = \frac{1}{3}NE_F \tag{5.15}$$

$$D(E) = \frac{dn}{dE} \tag{5.16}$$

$$D(E) = 2\frac{dn}{dE} \tag{5.17}$$

## Section 3: three-dimensional free electron gas and applications (5.2.2, 5.3)

# 5.2.2–5.3 Free electron gas in three dimensions and its applications (structured notes)

*Source: book-sources/bs-3-free-electron-theory.html, Puri and Babbar pp. 159–172. Equations 5.19 to 5.53 are cited by their book numbers. Figures are redrawn.*

### Before you start

- Volume of a sphere; counting states in $k$-space.
- Pressure from energy: $P = -\partial E/\partial V$; bulk modulus $B = -V\partial P/\partial V$.

> **PLUS TWO RECALL: VOLUME OF A SPHERE**
>
> **Recall.** $V_{\text{sphere}} = \frac{4}{3}\pi k^3$.
>
> **Re-derive.**
>
> 1. Each allowed $\mathbf{k}$ occupies $(2\pi/L)^3$.
> 2. Number of $\mathbf{k}$-points in the sphere: $\frac{4}{3}\pi k^3/(2\pi/L)^3$.
> 3. Times 2 for spin: $\frac{L^3}{3\pi^2}k^3$ (Eq. 5.34).
>
> **Saved.** Eq. (5.34).

### 5.2.2 Three-dimensional box

Schrödinger equation (5.19) in a cube of edge $L$; standing-wave solution (5.20), with travelling-wave form $\psi_{\mathbf{k}} = Ae^{i\mathbf{k}\cdot\mathbf{r}}$ (5.21). Periodic boundary conditions (5.23) fix

$$k_x = \frac{2\pi n_x}{L} \qquad k_y = \frac{2\pi n_y}{L} \qquad k_z = \frac{2\pi n_z}{L} \tag{5.24}$$

with $k^2 = 2mE_k/\hbar^2$ (5.22) and $E_k = (\hbar^2/2m)(k_x^2 + k_y^2 + k_z^2)$ (5.25). Normalised wave function (5.27): $\psi_{\mathbf{k}} = L^{-3/2}e^{i\mathbf{k}\cdot\mathbf{r}}$. The state is labelled by $k_x, k_y, k_z, m_s$.

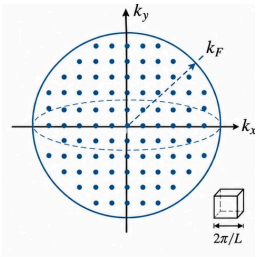



*Fig. 5.8 (book): sphere of radius $k_F$ in k-space containing the occupied states of N free electrons in the ground state.*

### 5.2.2.1 Filling of levels; Fermi sphere

At 0 K the occupied states fill a sphere of radius $k_F$ (Fig. 5.8). Counting (5.34): $\frac{L^3}{3\pi^2}k_F^3 = N$. Then

$$k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.35}$$

$$E_F = \frac{\hbar^2}{2m}\left(\frac{3\pi^2 N}{V}\right)^{2/3} \tag{5.36}$$

$$N = \frac{V}{3\pi^2}\left(\frac{2mE_F}{\hbar^2}\right)^{3/2} \tag{5.37}$$

$$v_F = \frac{\hbar}{m}\left(\frac{3\pi^2 N}{V}\right)^{1/3} \tag{5.38}$$

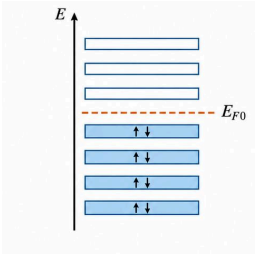



*Fig. 5.6 (book): distribution of electrons among the levels of a three-dimensional free electron model at 0 K.*

### 5.2.2.2 Density of states

From (5.39) and (5.37), $D(E)$ is parabolic in $E^{1/2}$:

$$D(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}E^{1/2} \tag{5.40}$$

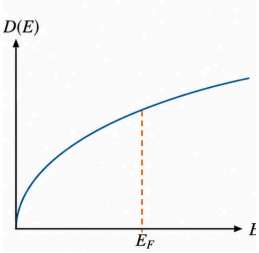



*Fig. 5.9 (book): variation of density of electronic states with energy for a free electron gas in three dimensions.*

$$D(E_F) = \frac{dN}{dE_F} = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{1/2} \tag{5.41}$$

$$\frac{dN}{N} = \frac{3}{2}\frac{dE_F}{E_F} \tag{5.42}$$

Hence $D(E_F)$ is of the order of $N/E_F$.

### 5.2.2.3 Fermi function

$$N(E) = D(E) f(E) \tag{5.43}$$

$$N(E)\,dE = D(E) f(E)\,dE \tag{5.44}$$

$$f(E) = \frac{1}{\exp\left[\dfrac{E - E_F}{k_B T}\right] + 1} \tag{5.28}$$

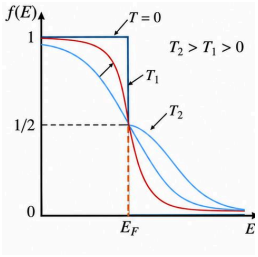



*Fig. 5.7 (book): Fermi distribution function at different temperatures; the variation of the Fermi energy with temperature is also shown.*

At 0 K: step function (5.29). At $E_F$: $f = \frac{1}{2}$. Shift of $E_F$ with $T$ (5.30): $E_F \cong E_{F_0}[1 - (\pi^2/12)(k_B T/E_{F_0})^2]$. Only $\sim k_B T$ near $E_F$ is affected.

$$E_F \cong E_{F_0}\left[1 - \frac{\pi^2}{12}\left(\frac{k_B T}{E_{F_0}}\right)^2\right] \tag{5.30}$$

### 5.2.2.4 Parameters at 0 K

$$E_{F0} = \frac{\hbar^2}{2m}(3\pi^2 n)^{2/3} \tag{5.47}$$

Average kinetic energy: $E_0 = \frac{1}{N}\int_0^{E_{F0}} E\,D(E)\,dE = \frac{3}{5}E_{F0}$.

$$E_0 = \tfrac{3}{5}E_{F0}$$

## 5.3 Applications

### 5.3.1 Electronic specific heat

Classical: $E = \frac{3}{2}Nk_BT$, $C_V = \frac{3}{2}Nk_B$ (5.49–5.50), about $3$ cal g-atom$^{-1}$K$^{-1}$ electronic, contrary to observation (electronic part at most 1% at room temperature). Quantum: only electrons within $k_BT$ of $E_F$ contribute, a fraction $T/T_F$; the printed result is

$$C_V \cong \frac{3}{2}Nk_B\left(\frac{T}{T_F}\right) \tag{5.51}$$

Proportional to $T$; zero at $T = 0$; with $T = 300$ K and $T_F = 3000$ K, about 0.01 of the classical value.

### 5.3.2 Examples 5.1 and 5.2 (book)

Example 5.1: sodium, $r = 1.86$ Å, bcc, $N/V = 2.52 \times 10^{28}$ m$^{-3}$, $E_{F0} = 3.11$ eV. Example 5.2: $P = \frac{2}{5}NE_{F0}/V$ (5.53) and the pressure–volume law (5.52); $B = \frac{5}{3}P$.

> **RESULT**
>
> 3D gas: $k_F$ (5.35), $E_F$ (5.36), $D(E)$ (5.40), $E_0 = \frac{3}{5}E_F$, $C_V \cong \frac{3}{2}Nk_B(T/T_F)$ (5.51).

## Summary

- (5.35)–(5.40) Fermi sphere and density of states; (5.28), (5.30) occupation; (5.51) specific heat.

## Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\nabla^2\psi_k(\mathbf{r}) + \frac{2m}{\hbar^2}E_k\psi_k(\mathbf{r}) = 0 \tag{5.19}$$

$$\psi_k(\mathbf{r}) = \sqrt{\frac{8}{L^3}}\sin\left(\frac{\pi n_x x}{L}\right)\sin\left(\frac{\pi n_y y}{L}\right)\sin\left(\frac{\pi n_z z}{L}\right) \tag{5.20}$$

$$\psi_k(\mathbf{r}) = A\,e^{i\mathbf{k}\cdot\mathbf{r}} = Ae^{i(k_x x + k_y y + k_z z)} \tag{5.21}$$

$$k^2 = \frac{2m}{\hbar^2}E_k \tag{5.22}$$

$$\psi(x+L,y,z) = \psi(x,y,z)\quad \psi(x,y+L,z) = \psi(x,y,z)\quad \psi(x,y,z+L) = \psi(x,y,z) \tag{5.23}$$

$$E_k = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2) \tag{5.25}$$

# 6.1–6.3 Band theory: the Bloch theorem and the Kronig–Penney model (structured notes)

*Source: book-sources/bs-4-band-theory.html, Puri and Babbar pp. 175–186. Equations 6.1 to 6.47 are cited by their book numbers. Figures are redrawn.*

## Before you start

- Periodicity $V(x+a)=V(x)$; plane waves; Bragg condition (box below).

> **PLUS TWO RECALL: CONSTRUCTIVE REFLECTION FROM PERIODIC PLANES**
>
> **Recall.** Reflection from periodic planes is constructive when the path difference equals $n\lambda$.
>
> **Re-derive.**
>
> 1. At normal incidence the condition is $n\lambda = 2a$.
> 2. With $k = 2\pi/\lambda$, this is $k = n\pi/a$.
>
> **Saved.** Eqs. (6.41)–(6.42).

## 6.1 Introduction

The constant-potential model fails for conductors versus insulators, the resistivity of insulators and semiconductors. The periodic potential of the ion cores (period $a$, Fig. 6.1) gives two results: (a) allowed bands separated by gaps; (b) $E(\mathbf{k})$ periodic in $\mathbf{k}$.

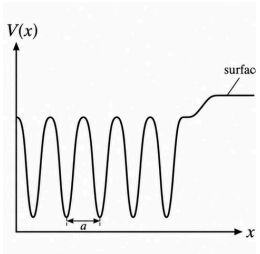



*Fig. 6.1 (book): one-dimensional periodic potential experienced by an electron in a crystal with lattice constant a; the potential near the surface is at the right.*

## 6.2 Bloch theorem

Constant potential: $\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E-V_0)\psi = 0$ (6.1); solutions $e^{\pm ikx}$ (6.2). Periodic potential (6.3). Bloch's theorem:

$$\psi(x) = e^{ikx}u_k(x) \qquad u_k(x) = u_k(x+a) \qquad (6.4\text{–}6.5)$$

Proof: the general solution is $Af(x)+Bg(x)$ (6.6); the translated functions are combinations (6.7); the eigenvalue condition (6.9–6.12); the Wronskian gives $\alpha_1\beta_2 - \alpha_2\beta_1 = 1$ (6.13); so $\lambda^2 - (\alpha_1+\beta_2)\lambda + 1 = 0$ (6.14).

$$\lambda^2 - (\alpha_1+\beta_2)\lambda + 1 = 0 \qquad (6.14)$$

**Case I** $(\alpha_1+\beta_2)^2 < 4$: $\lambda = e^{\pm ika}$ (6.15), $\psi(x+a) = e^{ika}\psi(x)$ (6.17): the Bloch property; bounded; allowed bands. **Case II** $(\alpha_1+\beta_2)^2 > 4$: $\lambda = e^{\pm\mu a}$ (6.18), unbounded; forbidden gaps.

3D: $\psi_{\mathbf{k}} = e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r})$ (6.20).

## 6.3 Kronig–Penney model

Square wells of width $a$, barriers $V_0$ of width $b$. $\alpha^2 = 2mE/\hbar^2$, $\beta^2 = 2m(V_0-E)/\hbar^2$ (6.23–6.25).

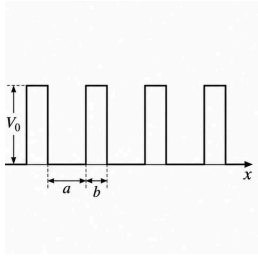



*Fig. 6.2 (book): one-dimensional periodic potential used by Kronig and Penney.*

$$\frac{\beta^2-\alpha^2}{2\alpha\beta}\sinh\beta b\,\sin\alpha a + \cosh\beta b\,\cos\alpha a = \cos k(a+b) \qquad (6.3$$

Delta-function limit: $V_0\to\infty$, $b\to 0$, $V_0 b$ fixed. Then (6.37), with $P = mV_0 ba/\hbar^2$ (6.38):

$$P\frac{\sin\alpha a}{\alpha a} + \cos\alpha a = \cos ka \qquad (6.39)$$

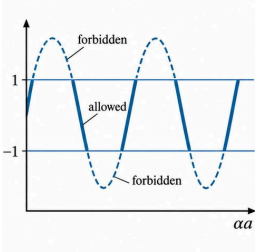



*Fig. 6.3 (book): left-hand side of Eq. (6.39) against αa for P = 3π/2; solid and broken lines on the axis are the allowed and forbidden regions.*

Allowed where the left side lies in $[-1, 1]$ (Fig. 6.3, $P = 3\pi/2$). Bands widen with energy and narrow with $P$. $P \to \infty$: $\alpha a = \pm n\pi$ (box levels). $P \to 0$: $\alpha = k$ (free electrons). Figure 6.4 shows the spectrum against $P$.

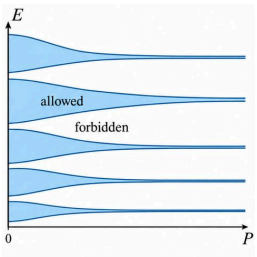



*Fig. 6.4 (book): allowed (shaded) and forbidden (open) energy ranges as a function of P.*

### 6.3.1 Zone schemes

Repeated zone; extended zone with discontinuities at $k = \pm n\pi/a$ (6.41), which give $n\lambda = 2a$ (6.42); reduced zone with $k = k' - G$ (6.43).

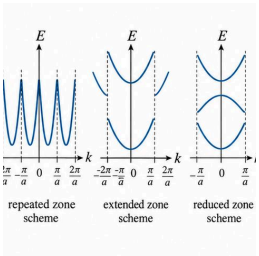



*Fig. 6.5 (book): E versus k in three zone schemes: (a) repeated, (b) extended, (c) reduced.*

### 6.3.2 States per band

$e^{ikL} = 1$ gives $k = 2\pi n/L$ (6.45–6.46); $dn = (L/2\pi)dk$ (6.47). The first zone holds $N$ states, so each band holds $N$ states and $2N$ electrons.

> **RESULT**
>
> Bloch functions (6.4), bands and gaps (6.15–6.18), Kronig–Penney condition (6.39), $N$ states per band (6.47).

## Summary

- Bands and gaps from periodicity; (6.4), (6.39), (6.41), (6.47).

## Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \tag{6.1}$$

$$\psi(x) = e^{\pm ikx} \tag{6.2}$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0 \tag{6.3}$$

$$\psi(x) = e^{\pm ikx}u_k(x) \tag{6.4}$$

$$u_k(x) = u_k(x + a) \tag{6.5}$$

$$\psi(x) = A\,f(x) + B\,g(x) \tag{6.6}$$

$$\begin{aligned} f(x + a) &= \alpha_1 f(x) + \alpha_2 g(x) \\ g(x + a) &= \beta_1 f(x) + \beta_2 g(x) \end{aligned} \tag{6.7}$$

$$\begin{aligned} \psi(x + a) &= A\,f(x + a) + B\,g(x + a) \\ &= (A\alpha_1 + B\beta_1)\,f(x) + (A\alpha_2 + B\beta_2)\,g(x) \end{aligned} \tag{6.8}$$

$$\left.\begin{aligned} A\alpha_1 + B\beta_1 &= \lambda A \\ \text{and}\qquad A\alpha_2 + B\beta_2 &= \lambda B \end{aligned}\right] \tag{6.9}$$

$$\psi(x + a) = \lambda\psi(x) \tag{6.10}$$

$$\begin{vmatrix} (\alpha_1 - \lambda) & \beta_1 \\ \alpha_2 & (\beta_2 - \lambda) \end{vmatrix} = 0 \tag{6.11}$$

$$\lambda^2 - (\alpha_1 + \beta_2)\lambda + \alpha_1\beta_2 - \alpha_2\beta_1 = 0 \tag{6.12}$$

$$\alpha_1\beta_2 - \alpha_2\beta_1 = 1 \tag{6.13}$$

$$\left.\begin{aligned} \lambda_1 &= e^{ika} \\ \text{and}\qquad \lambda_2 &= e^{-ika} \end{aligned}\right] \tag{6.15}$$

$$\left.\begin{aligned} \psi_1(x + a) &= e^{ika}\psi_1(x) \\ \text{and}\qquad \psi_2(x + a) &= e^{-ika}\psi_2(x) \end{aligned}\right] \tag{6.16}$$

$$\psi(x + a) = e^{\pm ika}\psi(x) \tag{6.17}$$

$$\lambda_1 = e^{\mu a} \qquad \text{and} \qquad \lambda_2 = e^{-\mu a} \tag{6.18}$$

$$\begin{aligned} \psi_1(x) &= e^{\mu x}u(x) \\ &\text{and} \\ \psi_2(x) &= e^{-\mu x}u(x) \end{aligned} \tag{6.19}$$

$$\psi_{\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{\mathbf{k}}(\mathbf{r}) \tag{6.20}$$

$$\begin{aligned} &\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0 \\ &\text{for } 0 < x < a \end{aligned} \tag{6.21}$$

$$\begin{aligned} &\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \\ &\text{for } -b < x < 0 \end{aligned} \tag{6.22}$$

# 6.4–6.5 Velocity, effective mass, and metals, insulators and semiconductors (structured notes)

*Source: book-sources/bs-4-band-theory.html, Puri and Babbar pp. 189–194. Equations 6.48 to 6.59 are cited by their book numbers. Figures are redrawn.*

## Before you start

- $F = ma$; chain rule of differentiation.

> **PLUS TWO RECALL: FORCE AND ACCELERATION**
>
> **Recall.** $F = ma$; $a = dv/dt$.
>
> **Re-derive.**
>
> 1. $v = (1/\hbar)\, dE/dk$ (6.49).
> 2. $a = (1/\hbar)(d^2E/dk^2)(dk/dt)$, and $\hbar\, dk/dt = e\mathcal{E}$ (6.52).
> 3. $a = (e\mathcal{E}/\hbar^2)\, d^2E/dk^2$ (6.53).
>
> **Saved.** Eq. (6.53).

## 6.4.1 Velocity of an electron

$$v = \frac{d\omega}{dk} = \frac{1}{\hbar}\frac{dE}{dk} \qquad (6.48\text{–}6.49)$$

Free electrons: $v \propto k$. In a band: $v = 0$ at $k = 0$ and $k = \pm\pi/a$; maximum at the inflexion $k_0$; falls beyond it (Fig. 6.6b).

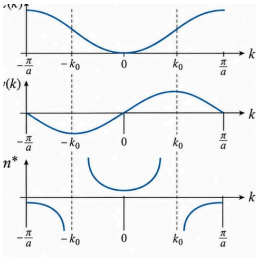



*Fig. 6.6 (book): energy, velocity and effective mass as functions of k; dashed lines mark the inflexion points.*

## 6.4.2 Effective mass

Crystal momentum rate of change equals the force (6.52). Acceleration (6.53), free electron (6.54), so

$$m^* = \frac{\hbar^2}{d^2E/dk^2} \qquad (6.55)$$

Positive in the lower half of the band, negative in the upper, infinite at inflexion points (Fig. 6.6c). Negative mass from Bragg reflection. Factor $f_k = m/m^*$ (6.56). Holes: positive charge and mass carriers in a nearly full band.

## 6.5 Metals, insulators, semiconductors

$$N_{\text{eff}} = \sum_k f_k \qquad (6.57)$$

$$N_{\text{eff}} = \frac{L}{\pi}\int_{-k_1}^{k_1} f_k\, dk \qquad (6.58)$$

$$N_{\text{eff}} = \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1} \qquad (6.59)$$

$N_{\text{eff}} = 0$ for a full band; maximum at the inflexion point.

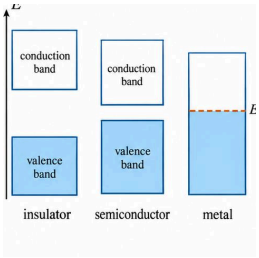



*Fig. 6.8 (book): electron distribution at 0 K in (a) an insulator, (b) an intrinsic semiconductor and (c) a metal.*

Insulator: wide gap (diamond about 6 eV). Intrinsic semiconductor: narrow gap (about 1 eV, Ge and Si); insulator at 0 K; conductivity rises with $T$. Metal: partly filled band; conductivity falls with $T$ (phonon scattering).

## Examples 6.1 and 6.2 (book)

Example 6.1: lowest band energy $E \approx \pi^2\hbar^2/2ma^2$ for $P \ll 1$ (as printed). Example 6.2: hole effective mass $+5.5 \times 10^{-30}$ kg, hole momentum $-1.05 \times 10^{-24}$ kg m s$^{-1}$, hole energy $10^{-19}$ J.

> **RESULT**
>
> $v = (1/\hbar)dE/dk$; $m^* = \hbar^2/(d^2E/dk^2)$; $N_{\text{eff}} = (2Lm/\pi\hbar^2)(dE/dk)_{k_1}$.

## Summary

- (6.49), (6.55), (6.59); full band has no conduction; metal, semiconductor, insulator distinguished by the gap and by $N_{\text{eff}}$.

## Book equations of this section not shown in the derivation above

Listed with their book numbers, as printed (from the verified transcription).

$$v = v_g = \frac{d\omega}{dk} \tag{6.48}$$

$$v = \frac{1}{\hbar}\frac{dE}{dk} \tag{6.49}$$

$$\begin{aligned} dE &= e\mathcal{E}dx \\ &= e\mathcal{E}vdt \\ &= \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt \end{aligned} \tag{6.50}$$

$$\frac{dk}{dt} = \frac{e\mathcal{E}}{\hbar} \tag{6.51}$$

$$\hbar\frac{dk}{dt} = \frac{dp}{dt} = F = e\mathcal{E} \tag{6.52}$$

$$a = \frac{e\mathcal{E}}{m} \tag{6.54}$$

$$f_k = \frac{m}{m^*} = \frac{m}{\hbar^2}\left(\frac{d^2E}{dk^2}\right) \tag{6.56}$$